AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Design basis: examples/sic-buck-400v-200v-5kw/design/design_decisions.json (schema 3, 2026-10-10) — single source of truth; electrical model automation/kicad_tools/sic_buck_power_model.py; generator automation/kicad_tools/generate_sic_schematics.py.
400 V (360–440 V) to 200 V / 25 A / 5 kW / 50 kHz synchronous buck; Wolfspeed C4MS025120K; TMS320F28335 control board on a stacked SELV mezzanine.
Net classes: HV_VIN, HV_SW, HV_PGND, HV_VOUT, DRV_HS, DRV_LS, DESAT_HS, DESAT_LS, SENSE_ISO, PGND_LV — HV classes 3 mm, driver / DESAT / sensing islands 0.3 mm intra-class, PE 5 mm, CTRL_SELV and Default 0.25 mm (70 um copper).
Board rules (.kicad_dru): HV side to CTRL_SELV 9 mm on all layers; HV to PE 5 mm; HV functional 3 mm outer / 1 mm inner; HV to board edge 1 mm; 39 pin-field exceptions derived from pad geometry (14 barrier, 17 HV, 8 package pitch) — creepage review.
Barrier parts (UCC21710 x2, UCC14141 x3, AMC1311 x2, TMCS1133) need the 1.2 mm non-plated isolation slot of the baseline; the custom AIPE footprints embed it, the UCC21710 library footprint does not (layout task).
TO-247-4 pad order 1 D, 2 S, 3 KS, 4 G is an ASSUMPTION (Wolfspeed PRD 05966 front view); unverified parts and placeholders are listed in components/bom_power.csv notes and in the model docstring.
Evidence: native ERC manifests under verification/erc/<run>/run.json; review PDF under review/; this drawing has no CAD evidence beyond those runs. AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED.

Input, half bridge, inductor, output, bleeders

G_H
G_L
ISENSE
OC_N
SW
VOC_SET
VOUT

文件: 01_power_stage.kicad_sch

High-side isolated gate driver and bias

BIAS_EN_F
FLT_H_N
G_H
PG_BIAS_H_N
PWM_H
PWM_H_F
PWM_L_F
RDY_H
RESET_N_F

文件: 02_gate_driver_hs.kicad_sch

Low-side isolated gate driver and bias

BIAS_EN_F
FLT_L_N
G_L
PG_BIAS_L_N
PWM_H_F
PWM_L
PWM_L_F
RDY_L
RESET_N_F
SW

文件: 03_gate_driver_ls.kicad_sch

Isolated voltage sensing island and its bias module

BIAS_EN_F
PG_BIAS_S_N
VIN_OUTN
VIN_OUTP
VOUT
VOUT_OUTN
VOUT_OUTP

文件: 04_sensing_island.kicad_sch

SELV strip: mezzanine, conditioning, relay and fan interface

FLT_H_N
FLT_L_N
ISENSE
OC_N
PG_BIAS_H_N
PG_BIAS_L_N
PG_BIAS_S_N
RDY_H
RDY_L
VIN_OUTN
VIN_OUTP
VOUT_OUTN
VOUT_OUTP

BIAS_EN_F
FAULT_N
PWM_H
PWM_L
RESET_N_F
SPARE_D1
SPARE_D2
VIN_SENSE
VOC_SET
VOUT_SENSE

文件: 05_selv_strip.kicad_sch

Heatsink envelope, terminals, mounting, test points

ISENSE
VIN_SENSE
VOUT_SENSE

FAULT_N
PWM_H_F
PWM_L_F
SPARE_D1
SPARE_D2

文件: 06_mechanical.kicad_sch

AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Sheet: /
File: AIPE_SiC_Buck_Power.kicad_sch

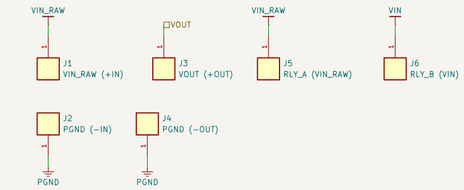
Title: SiC buck 400V / 200V / 5kW — power board

Size: A3 | Date: | **Rev: V2.2**
KiCad E.D.A. 10.0.7 | Id: 1/7

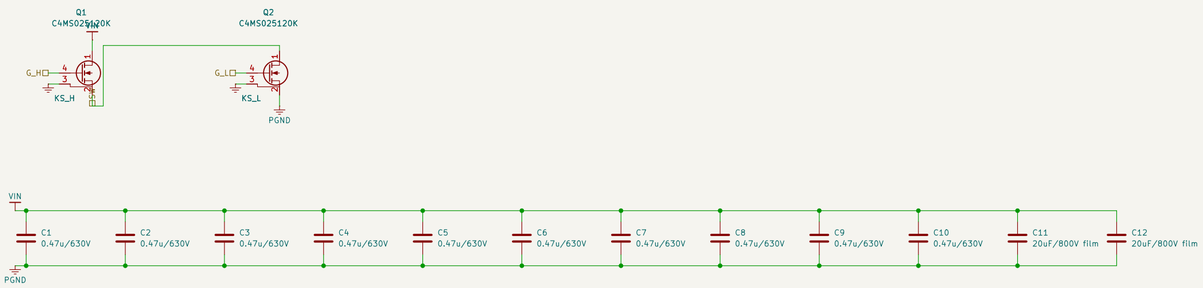
Input, half bridge, inductor, output, bleeders

Q1/Q2 pads 1, D, 2, S, 3, KS, 4, G follow the Wolfspeed PRD 05966 front-view lead order — ASSUMPTION, verify the footprint pad-1 side before fabrication. HV classes: VIN_RAW/VIN (HV_VIN), SW (HV_SW), VOUT_L/VOUT (HV_VOUT), PGND (HV_PGND) plus bleeder, indicator, ladder-tap, snubber and damping nodes. DNP: snubber, damping leg, MOV, PGND–PE link, Y2 capacitors.

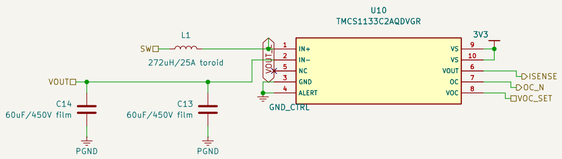
Power terminals, precharge PTC, MOV



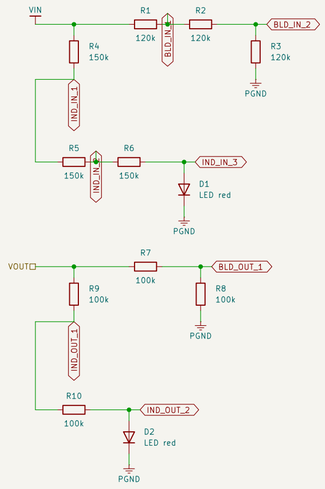
Half bridge, HF capacitor bank, DC-link capacitors



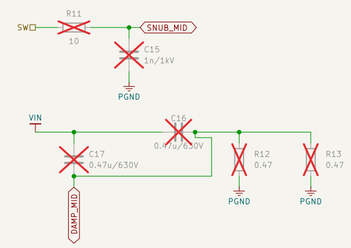
Output inductor, current sensor, output capacitors



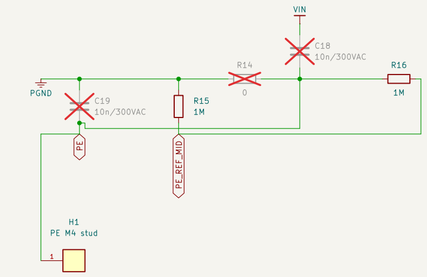
Bleeders and LIVE indicators



Snubber and damping leg (DNP)



PE network



Supply provenance (PWR_FLAG)



AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Sheet: /Input, half bridge, inductor, output, bleeders/
File: 01_power_stage.kicad_sch

Title: Input, half bridge, inductor, output, bleeders

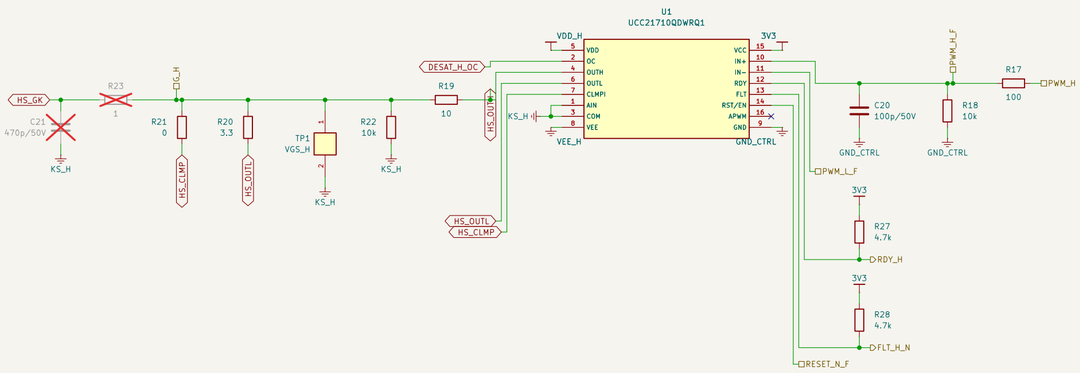
Size: A2 | Date: | Rev: V2.2

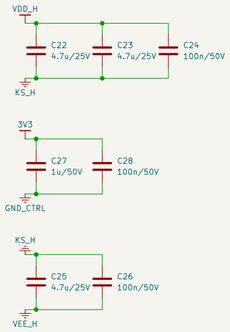
KiCad E.D.A. 10.0.7 | Id: 2/7

High-side isolated gate driver and bias

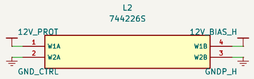
UCC21710 IN+ = PWM_H_F, IN- = PWM_L_F (hardware interlock KR-GD-009); AIN tied to COM (CR-01); APWM not connected. UCC14141 VDD-VEE 19.0 V (107 k / 16.2 k) and COM-VEE 4.0 V (9.76 k / 16.2 k) give +15 V / −4 V at the gate; RLIM 1 k is a placeholder (human review). Decoupling sits between the driver pins and the gate/Kelvin connection (KR-PCB-002); KS_H is never joined to SW on the PCB (KR-PCB-003).

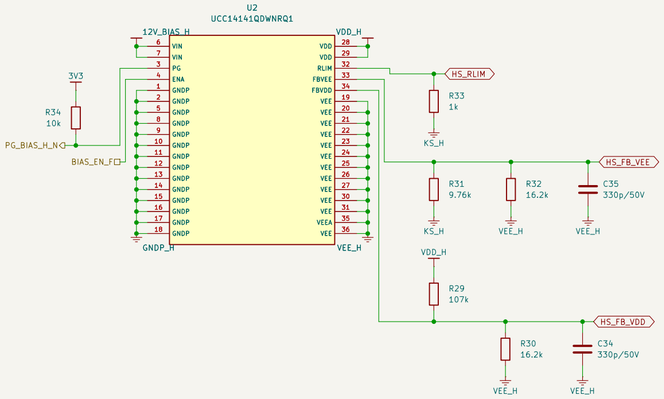
HS isolated gate driver (UCC21710-Q1): PWM input, gate network, status pull-ups, decoupling

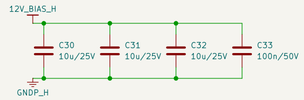


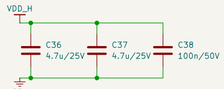


HS isolated bias supply (UCC14141-Q1)



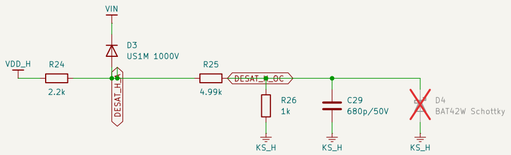








HS DESAT protection network



Supply provenance (PWR_FLAG)



AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Sheet: /High-side isolated gate driver and bias/
File: 02_gate_driver_hs.kicad_sch

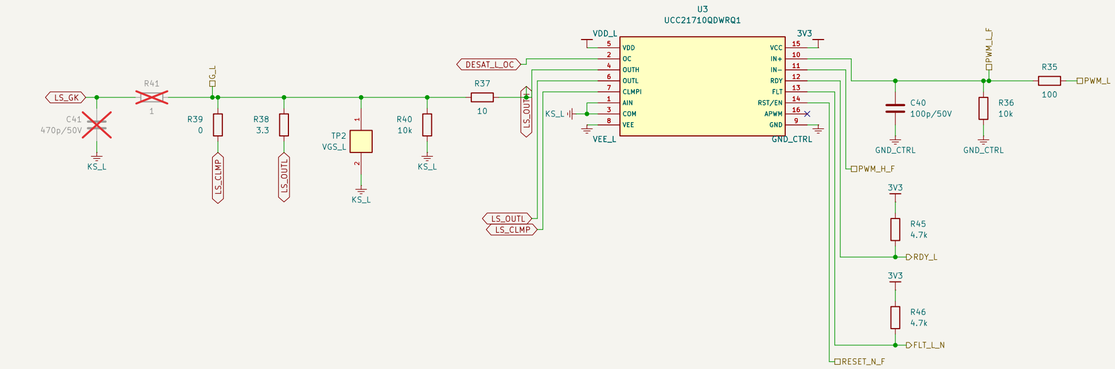
Title: High-side isolated gate driver and bias

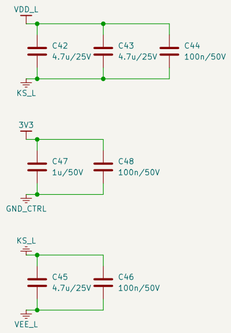
Size: A2 | Date: | **Rev: V2.2**
KiCad E.D.A. 10.0.7 | Id: 3/7

Low-side isolated gate driver and bias

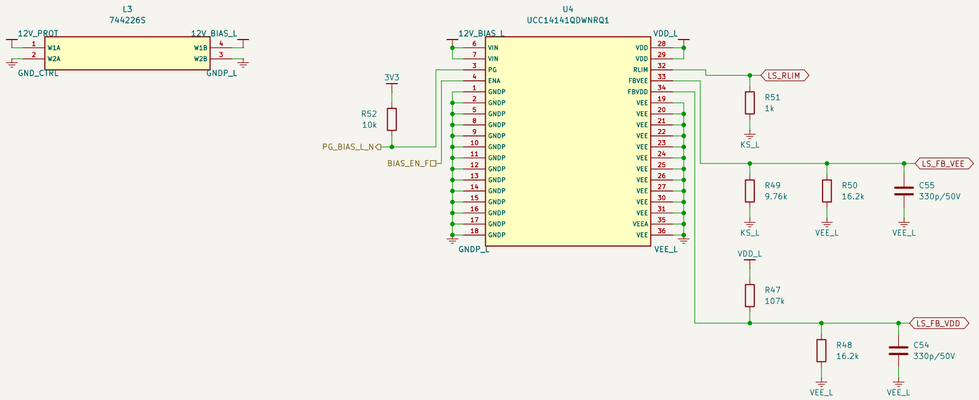
Mirror of the high-side channel: IN+ = PWM_L_F, IN- = PWM_H_F; DESAT diode cathode on SW; KS_L is never joined to PGND on the PCB (KR-PCB-003). Isolation slot under the library SOIC-16W footprint is a layout task.

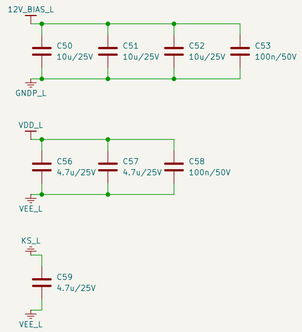
LS isolated gate driver (UCC21710-Q1): PWM input, gate network, status pull-ups, decoupling



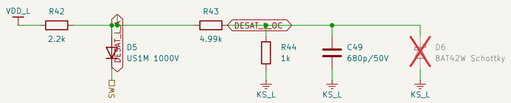


LS isolated bias supply (UCC14141-Q1)





LS DESAT protection network



Supply provenance (PWR_FLAG)



AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Sheet: /Low-side isolated gate driver and bias/
File: 03_gate_driver_ls.kicad_sch

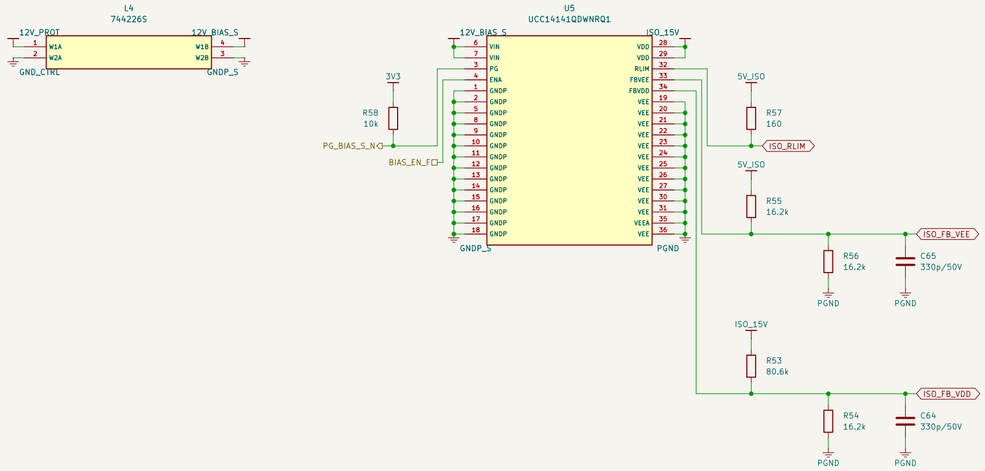
Title: Low-side isolated gate driver and bias

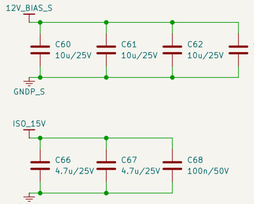
Size: A2 | Date: | **Rev: V2.2**
KiCad E.D.A. 10.0.7 | Id: 4/7

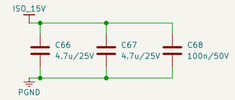
Isolated voltage sensing island and its bias module

Third UCC14141 (CR-12): VDD–VEE 15 V (80.6 k / 16.2 k, A–P2), COM–VEE 5.0 V = 5V_ISO, VEE = PGND. AMC1311 ladders 4 × 750 k (VIN, 602 V full scale) and 2 × 750 k (VOUT, 302 V) with 10.0 k bottoms; the taps VIN_DIV_T1…T3 and VOUT_DIV_T1 are HV nodes (classes HV_VIN / HV_VOUT, 120 V steps).

ISO isolated bias supply (UCC14141-Q1)

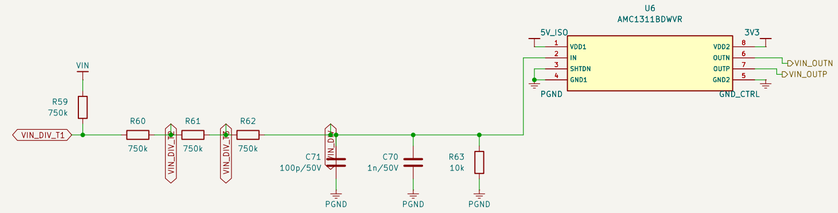


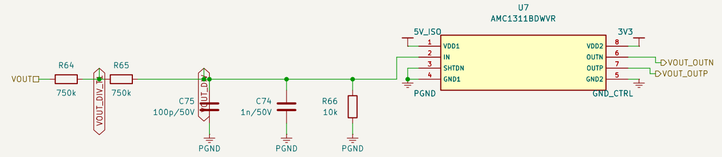


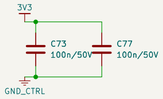


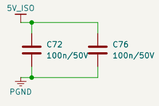


Isolated VIN / VOUT sensing (AMC1311)

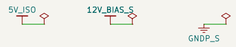








Supply provenance (PWR_FLAG)



AI-GENERATED ENGINEERING PROTOTYPE – V2.2 – HUMAN REVIEW REQUIRED

Sheet: /Isolated voltage sensing island and its bias module/
File: 04_sensing_island.kicad_sch

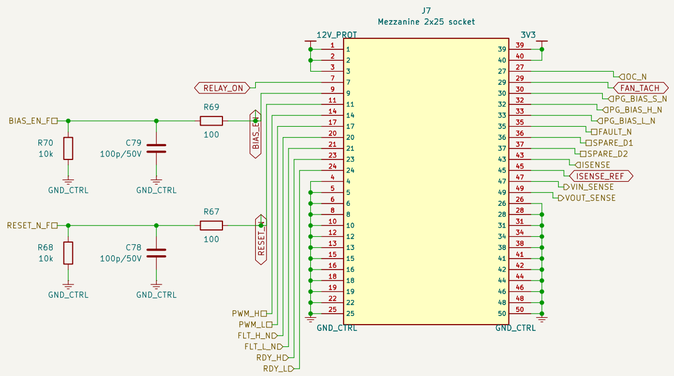
Title: Isolated voltage sensing island and its bias module

Size: A2 | Date: | **Rev: V2.2**
KiCad E.D.A. 10.0.7 | Id: 5/7

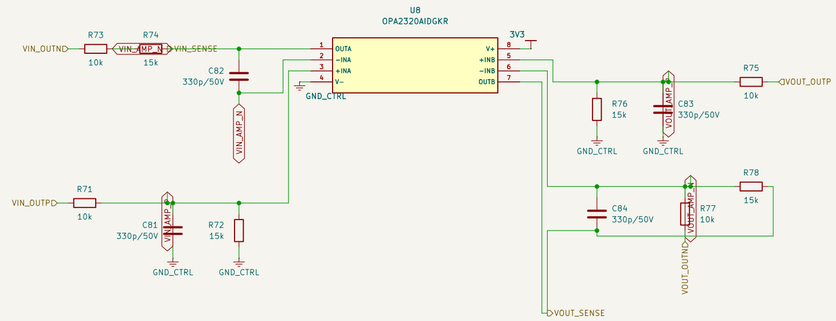
SELV strip: mezzanine, conditioning, relay and fan interface

Mezzanine socket SSW-125-01-G-D with the pin assignment of design_decisions.json: receiving networks 100 ohm / 100 pF / 10 k pull-down on RESET_N and BIAS_EN (A–P6); OPA2320 difference amplifiers gain 1.5; REF3125 → VOC_SET 0.750 V (I_OC 37.5 A); ISENSE_REF = 0.1 × 3V3 (A–P1); AO3400A + SS14 relay coil drive; fan header with tacho pull-up. All nets on this sheet are SELV (CTRL_SELV). PWR_FLAG symbols state supply provenance for ERC; they do not prove voltage, current capacity or protection.

Mezzanine socket and receiving networks



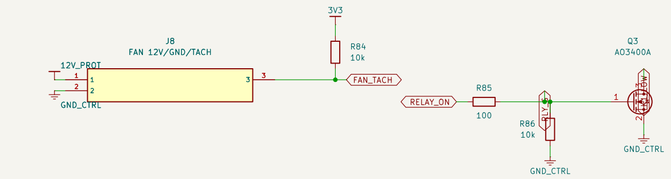
OPA2320 difference amplifiers



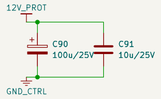


REF3125 reference, OC threshold, ISENSE_REF

Relay coil drive, fan header, 12 V bulk







Supply provenance (PWR_FLAG)



AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Sheet: /SELV strip: mezzanine, conditioning, relay and fan interface/
File: 05_selv_strip.kicad_sch

Title: SELV strip: mezzanine, conditioning, relay and fan interface

Size: A2 | Date: | Rev: V2.2
KiCad E.D.A. 10.0.7 | Id: 6/7

Heatsink envelope, terminals, mounting, test points

Heatsink envelope LAM 5 D K 100 12 (corner holes are placeholders), 6 x M4 baseplate standoffs, 4 x M3 control-board standoffs, bare-PCB test pads. The PE stud and PGND-PE network are on sheet 01.

Heatsink envelope and mounting holes

HS1
LAM 5 D K 100 12

H2
M4 standoff

H3
M4 standoff

H4
M4 standoff

H5
M4 standoff

H6
M4 standoff

H7
M4 standoff

H8
M3 standoff

H9
M3 standoff

H10
M3 standoff

H11
M3 standoff

Bare-PCB test pads (SELV signals)

3V3
TP3
3V3

PWM_H_F
TP6
PWM_H_F

PWM_L_F
TP7
PWM_L_F

FAULT_N
TP8
FAULT_N

ISENSE
TP9
ISENSE

VIN_SENSE
TP10
VIN_SENSE

VOUT_SENSE
TP11
VOUT_SENSE

SPARE_D1
TP12
SPARE_D1

SPARE_D2
TP13
SPARE_D2

TP4
GND_CTRL
GND_CTRL

TP5
GND_CTRL
GND_CTRL

AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Sheet: /Heatsink envelope, terminals, mounting, test points/
File: 06_mechanical.kicad_sch

Title: Heatsink envelope, terminals, mounting, test points

Size: A3	Date:	**Rev: V2.2**
KiCad E.D.A. 10.0.7		Id: 7/7